

Cisco Secure Workload - NIST SP 800-171 Compliance Report

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Customer-Facing Compliance Report

Framework: NIST SP 800-171 Revision 3 - Protecting Controlled Unclassified Information (CUI)

Primary audience: Defense industrial base contractors, federal contractors and subcontractors, and any organisation processing or storing CUI on non-federal systems.

CSW positioning: CUI enclave isolation, access enforcement, system and communications protection (03.01 and 03.13 families), and CMMC Level 2 underpinning evidence.

Framework Version Note

References NIST SP 800-171 Rev. 3 (published May 2024). Rev. 3 restructured the families; common family abbreviations used here (e.g. 03.01 Access Control, 03.13 System and Communications Protection) follow the Rev. 3 numbering. Validate against the version applicable to the customer's contract (Rev. 2 numbering may still be referenced in legacy contracts).

CSW UI Navigation Note

Cisco does not yet publish framework-specific CSW UI navigation for this standard; framework-specific navigation is on the product roadmap. This report references CSW *capabilities* and the *evidence artifacts* they produce. The reference design (CSW-NIST-800-171-Technical-Reference design.md) in this folder contains the deeper engineering detail.

How to read the Coverage column

Every report in this set uses the same three coverage words. Coverage describes **what evidence CSW produces** for a topic — it never means the control is passed.

Direct — Secure Workload produces the primary evidence artifact for that topic (segmentation, policy, flow/process telemetry, inventory, vulnerability + reachability, forensic reconstruction). Direct is still evidence, not an attestation.

Supporting — Secure Workload contributes an input that feeds or complements another control, tool, or process (e.g. detection input only, reconciled against an external register, one log source among several).

Evidence Required — Secure Workload has no record for this topic. The customer supplies the contract, the analysis, or the HR / physical-security evidence from another control.

The Coverage ratings in the topic map below are machine-derived from each row's wording and are flagged for SE review before customer use.

Executive Summary

Cisco Secure Workload (CSW) supports a defined subset of this framework. The strongest customer story is not that CSW “certifies” compliance; it is that CSW turns workload communication and inventory into evidence that the customer, the customer’s compliance team, and the customer’s assessor can review. This report describes which framework topics CSW supports, which it does not, what evidence artifacts CSW produces, and how often the customer should collect them. CSW does **not** replace the customer’s governance, policy, or assessor judgement, and this report is **not** a control attestation.

Suggested CSW Scope Pattern

NIST-800-171

- └─ CUI-Enclave (workloads processing or storing CUI)
- └─ CUI-Adjacent-Services (identity, logging, patching adjacent to CUI)
- └─ Admin-Paths (jump hosts)
- └─ External-Connections (CUI-related external paths)

This scope pattern should be validated with the customer’s architecture, application owners, compliance team, and assessor before it is used for formal evidence.

In Scope for CSW Evidence

- 03.01 Access Control (workload-to-workload allowlist)
- 03.13 System and Communications Protection (workload-to-workload segmentation)
- 03.04 Configuration Management (workspace baseline)
- 03.11 Risk Assessment (reachability-weighted vulnerability evidence)
- 03.14 System and Information Integrity (monitoring)
- 03.06 Incident Response (forensic input)

Out of Scope (Must Be Evidenced by Other Controls / Tools / Processes)

- 03.05 Identification and Authentication (identity layer)
- 03.09 Personnel Security
- 03.10 Physical Protection
- 03.08 Media Protection

- 03.13 Cryptography enforcement (CSW detects plaintext flows but does not enforce encryption)
- CMMC assessment scoring

Framework Topic → CSW Capability Map

Framework topic	Coverage	CSW capability	Evidence artifact
03.01 Access Control (workload layer)	Direct	Policy workspace + enforcement / violation log	Workspace snapshot + violation log
03.13 System and Communications Protection	Direct	Default-deny at CUI enclave boundary	Workspace snapshot showing boundary deny rules
03.04 Configuration Management	Direct	Workspace baseline and change log	Change-controlled policy workspace export
03.11 Risk Assessment (reachability)	Direct	Reachability-weighted CVE evidence	Prioritised remediation list
03.14 System and Information Integrity	Direct	Continuous flow + process telemetry; alerts	Alert and anomaly log scoped to CUI-Enclave
03.06 Incident Response	Direct	Forensic flow + process search	Investigation pack with timeline and impact scope

Evidence Collection Approach

The table below describes per-topic evidence collection - what to collect, the CSW source area, and a suggested cadence. The cadence should be tuned to the customer's audit cycle, regulatory cadence, and risk appetite.

Topic	What to collect	CSW source	Suggested cadence
CUI enclave inventory	Workloads processing or storing CUI	Inventory + scope membership export filtered by data_class=cui	Monthly
Enclave boundary evidence	Workspace snapshot + observed-vs-approved flow comparison	Policy workspace export + ADM export	Quarterly + after each change
External connection summary	Outbound flow from CUI-Enclave	Flow search filtered to CUI-Enclave egress	Monthly

Topic	What to collect	CSW source	Suggested cadence
Vulnerability + reachability	CVE list weighted by reach into CUI-Enclave	Vulnerability report scoped to NIST-800-171	Monthly
Incident sample	One representative incident timeline	Flow + process search	Per incident

POV / Workshop Approach

1. Confirm the framework version and assessment cycle with the customer's compliance team.
2. Select one business service, regulated boundary, or critical workload group to start with; do not attempt to instrument the entire estate in the POV.
3. Validate workload coverage and install sensors or configure supported connectors. Document the cannot-instrument register; this is itself evidence.
4. Apply labels for application, environment, owner, data class, criticality, compliance scope, and any framework-specific tags (e.g. `tenant`, `prototype`, `isa_level`, `npi`, `cui`, `ephi`).
5. Observe flows over a representative business window. Minimum two weeks; longer if the customer has month-end, quarterly, or seasonal cycles relevant to the scope.
6. Use ADM to generate a candidate policy. Review with the application owner. Run in Simulation mode for at least two weeks before any enforcement.
7. Package inventory, scope, flows, policy, exceptions, and known gaps as the evidence output. Cross-reference this report's evidence collection table with the customer's existing audit work-papers.

Boundaries and Complementary Controls

CSW does not replace CUI marking, federal contract requirements (DFARS 252.204-7012, FAR 52.204-21, etc.), System Security Plan ownership, POA&M ownership, identity governance, media protection, encryption (FIPS-validated modules), personnel security, or CMMC assessment.

Assessor / Auditor Caveat

CSW provides technical evidence inputs. The customer's qualified assessor, auditor, certifying body, or regulator determines compliance status. Validate all framework-topic mappings, control IDs, paragraph references, and section numbers in this report against the current official framework text. If the customer is being assessed against a version other than the one referenced in the framework version note above, revalidate before use.

Disclaimer

This report is for informational and planning purposes. It is not legal, regulatory, audit, or certification advice. Validate all mappings against the current official framework text, the customer's environment, and qualified compliance, legal, and audit professionals.